

AMIS 1012

Ethics in Computing



Chapter 7: Software Development

Chapter Outline

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1. Software development methodology
2. Software testing
3. Quality assurance and standards
4. Software quality and assurance techniques





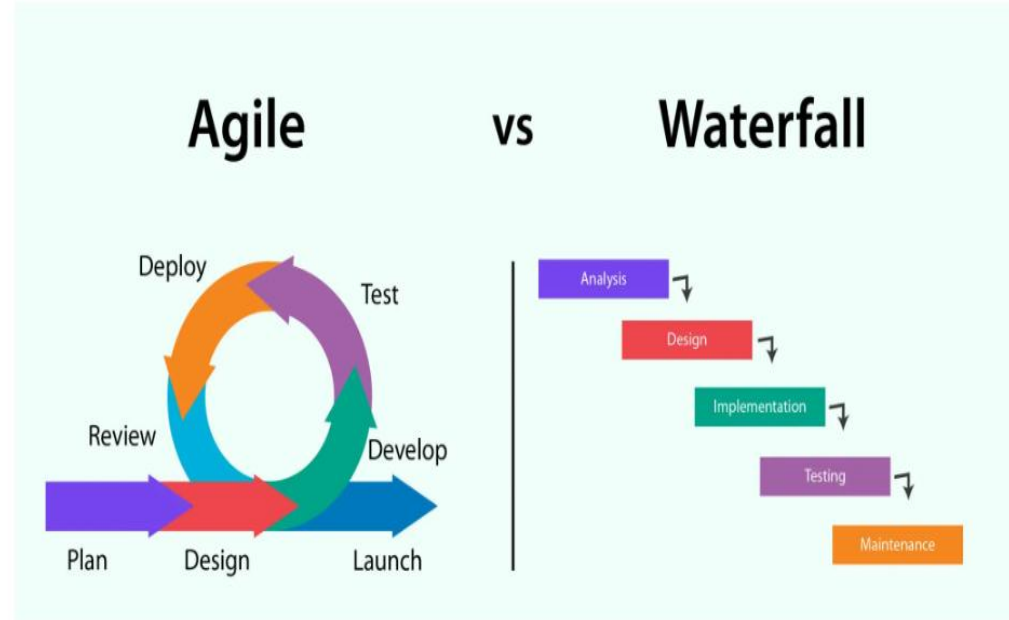
1. Software development methodology

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- Software development methodology is a **tried- and-true work procedure** that helps system analysts, designers, project managers, and others **to produce high-quality software** in a **managed** and **orderly** manner.
- The tasks in the software development phase, as well as person and community obligations for completing them, are described by a **technique**.
- There are specific techniques for carrying out the various functions, such as recording the logic of a computer application using a flowchart.

1. Software development methodology

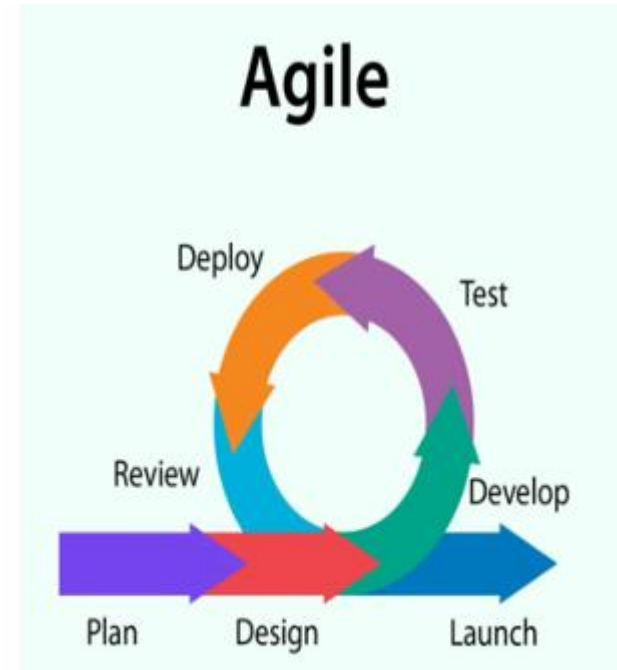
- Examples of Software Development Methodology
 - Agile
 - Waterfall



Source: <https://www.zaigoinfotech.com/guide-to-agile-software-development-methodology/>

1. Software development methodology

- Agile Software Development Methodology
 - flexible, iterative approach
 - emphasizes continuous delivery, customer feedback, and adapting to change
 - breaks the project into small, manageable portions
 - allowing teams to adjust direction based on client feedback and project evolution
 - emphasis on delivering functional product increments rapidly and efficiently

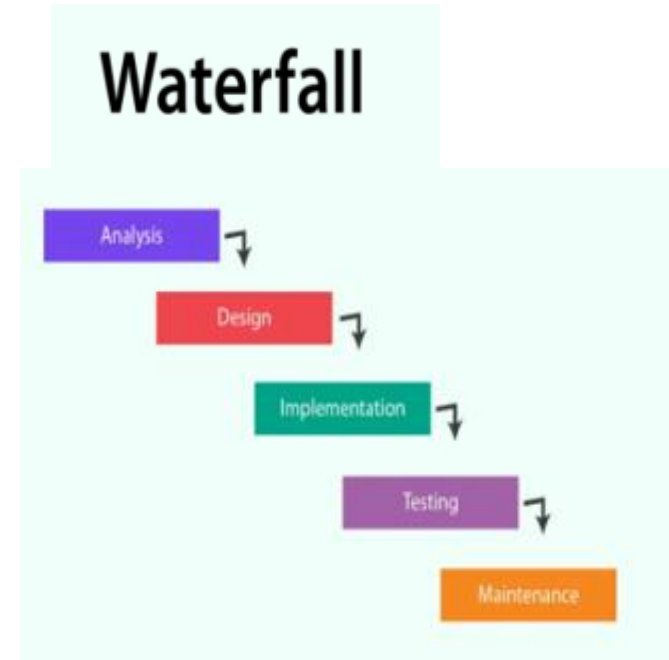


Source: <https://www.zaigoinfotech.com/guide-to-agile-software-development-methodology/>

1. Software development methodology

Waterfall Software Development Methodology

- traditional, linear approach
- a strict sequence of phases: requirements, design, implementation, testing, deployment, and maintenance
- Each phase must be completed before moving on to the next, with little room for revisiting earlier phases
- suits projects with well-defined requirements and where changes are unlikely or costly to implement after the project starts





1. Software development methodology

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- Identifying and eliminating a defect early in the software development process (requirements definition) will cost up to 100 times less than removing a defect in a piece of software that has already been shipped to consumers.





1. Software development methodology

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- If a flaw is discovered later in the production process, any **rework** on the deliverables created earlier in the process would be required.
- Thus, the **costs of communicating** and **correcting the mistake** would be higher.
- As a result, most software engineers strive to find and **eliminate mistakes early in the development process**, not just to save money but also to **increase software quality**.



1. Software development methodology

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- A product warranty lawsuit may be filed if it has underlying flaws that cause damage to the consumer.
- Failing to carefully and reliably design applications will result in significant liability exposure.



2. Software testing

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- Software testing is the method of **evaluating the functionality of a software program** with the goal of determining whether the designed software **meets the required specifications** and **identifying bugs** in order to deliver a better product.



2. Software testing

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- There are several types of testing methods, such as:
 - **Manual testing**
 - Process of **manually evaluating software** in order to learn more about it and determine what is working and what is not.
 - **Verify all of the functionality listed in the specification documents**, but it also often entails testers putting the software through its paces from the perspective of its end customers.



2. Software testing

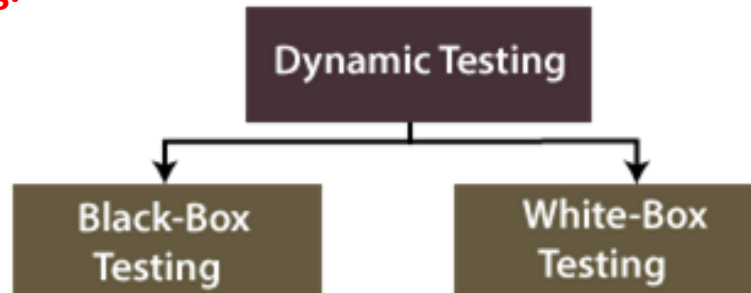
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- There are several types of testing methods, such as:
 - **Automation Testing**
 - Method of finding bugs in software by **using an automated tool**.
 - Using automation software, **testers run test scripts and produce test results** automatically in this process.



2. Software testing

- There are several types of testing methods, such as:
 - **Dynamic Testing**
 - Code for a finished unit of software is tested by **directly entering test data** and **matching the results with the predicted results**.
 - Dynamic testing can be divided into two types: **black-box testing and white-box testing**.



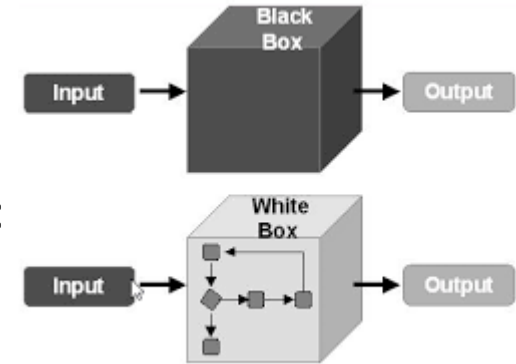
2. Software testing

- There are several types of testing methods, such as:

- Dynamic Testing

- Black-box testing**

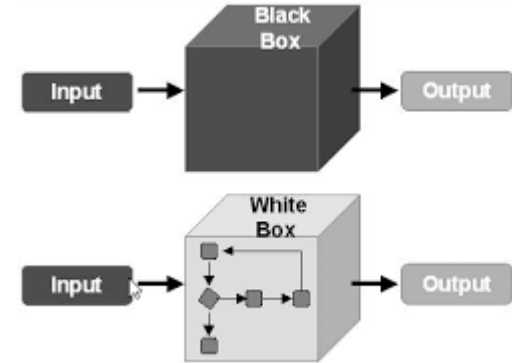
- Software unit as a system **with predicted input and output actions** but **unexplained internal workings** (a black box).
 - The unit passes the test if it **exhibits the predicted behaviors** with any of the input data in the test suite.
 - Black-box checking is done **without the tester knowing something about the code's composition or design**.
 - As a result, it is often **performed by someone other than the code author**.



2. Software testing

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- There are several types of testing methods, such as:
 - Dynamic Testing
 - **White-box testing**
 - Considers the software machine as a system **with planned input and output behaviors** and **known internal workings**.
 - White-box research entails thoroughly **understanding the logic of the program unit and testing all feasible logic pathways** through it.
 - Each program statement must be executed at least once, so the test data must be carefully assembled.





3. Quality assurance and standards

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- **Quality assurance (QA)** refers to approaches used during the production period to **ensure a product's consistent service**.
- Quality is difficult to define, but it can be summarized as “fit for use or purpose.” It's all about **satisfying consumers' desires** and desires in terms of product **functionality, style, efficiency, longevity, and price**.





3. Quality assurance and standards

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- Assurance is an optimistic statement about a good or service that inspires confidence. It is the assurance that a **good or service can perform well**. It ensures that the product will perform flawlessly and in accordance with the standards or specifications.
- Software testing QA is defined as a **method for assuring the quality of software goods or services** that a business provides to its customers.

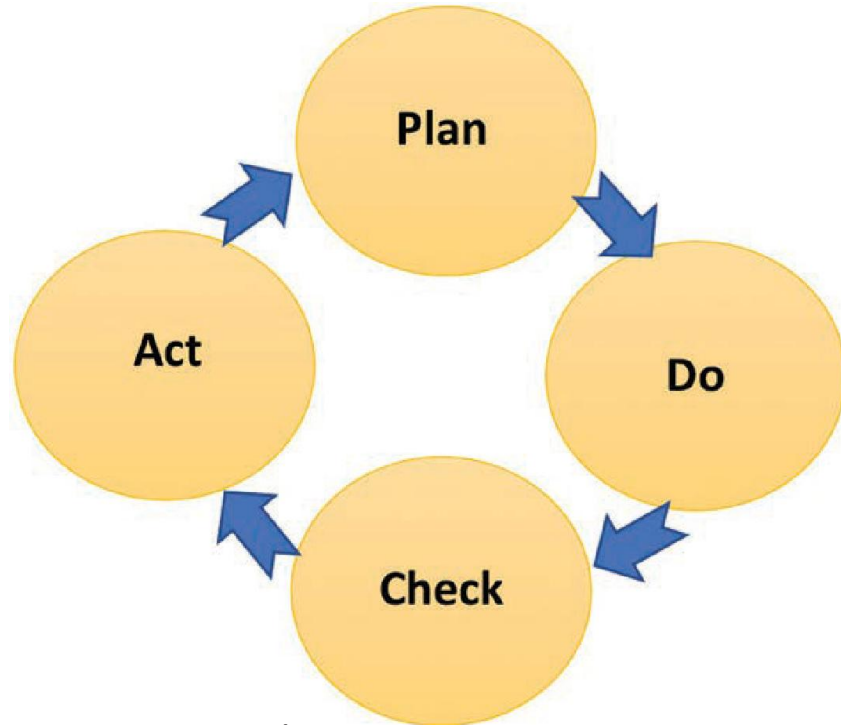


3. Quality assurance and standards

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- QA is concerned with making the software development process more **dependable** and **competitive** while adhering to the quality standards set for software products.
- In QA, the **PDCA cycle**, commonly known as the Deming cycle, is a fixed cycle. Plan, do, check, and act are the four stages of this cycle.
- These processes are **repeated on a regular basis** to guarantee that the **organization's operations are evaluated and improved**.

3. Quality assurance and standards



Quality assurance process

- **Plan** and create process-related objectives, as well as the processes required to deliver a high-quality final result.
- **Do**: Implement the plan. Develop and test processes, as well as “do” system changes
- **Check**: Procedures are monitored, and checked to confirm whether they meet their objectives.
- **Act**: A QA tester may take the necessary measures to improve processes.

3. Quality assurance and standards

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- An organization uses QA **to verify that the product is developed and delivered properly.** This helps to **reduce the number of problems and faults in the final product.**



4. Software quality and assurance techniques

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- A high-quality product **fulfills the expectations of its consumers**. The suitability of software items for usage is typically described in terms of fulfilling the Software Requirements Specifications (SRS) guide's requirements.



4. Software quality and assurance techniques

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- Many quality methods are included in today's knowledge of software product quality, including the following:
 - **Scalable** – able to operate on a range of operating systems, on various computers, or with other software devices.
 - **Usability**: more usable if all types of consumers can conveniently invoke the system's features.
 - **Reusability**: separate components of a software system can be easily replicated to produce new applications.

4. Software quality and assurance techniques

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- Many quality methods are included in today's knowledge of software product quality, including the following (cont.):
 - **Correctness**: SRS document's different specifications have been applied properly.
 - **Maintainability**: glitches can be easily fixed when they arise, new tasks can be easily applied to the product, and the product's functionalities can be easily changed, among other things.

4. Software quality and assurance techniques

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- A quality control scheme is one of the most common ways for businesses to ensure that the goods they produce are of the highest possible quality.
- **Individual responsibilities and managerial structure are the component of a quality system:**
 - The entire organization is responsible for a quality scheme. Any organization may have several quality departments that conduct different quality system activities.
 - The arrangement's quality scheme should have the **approval of the top management**. Some employees would **take the quality system negatively if the quality system is not supported at a high level in the business**.

4. Software quality and assurance techniques

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- Activities in the quality management system:
 - The functions of the quality system include the following:
 - Project auditing
 - Quality system review
 - Development of principles, procedures, and protocols, among other things
 - Production of publications for senior management summarizing the success of the organization's quality system.



Question?

"Ethics is knowing the difference between what you have a right to do and what is right to do." - Potter Stewart



Disclaimer:

The slide content has been sourced from the following reference books and Internet resources. Refer to the following list for more details and further reading:

1. Awari, G.K., & Warjekar, S.V. (2022). *Ethics in Information Technology: A Practical Guide*. CRC Press, Taylor & Francis Group.
2. ZAIGO Infotech. (2024). The Ultimate Guide to Agile Software Development Methodology.
<https://www.zaigoinfotech.com/guide-to-agile-software-development-methodology/>

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